

RESCUE JUNIOR - Bill of Materials (BOM)

Team name: DELTA

Every Line/Maze team has to submit a bill of materials for their robot

Instructions:

*Enter Team name above
*All costs need to be in local currency and their approximate conversion into dollars.
*All components worth less can be summarized in one line quantity 1 and their total cost (e.g. Screws)
*In the Local Currency Name column, the team must enter the name of the local currency, if it is different from US dollars.
*In the software sheet, in the Author column, the name of the designer, writer, or company that owns the software or library should be entered. If the team created their own algorithm, dataset, or AI model, they can include their names as authors.
*On the hardware sheet, in the "Author" column, enter the name of the company, engineer, or part manufacturer. If the team made a custom part, such as the robot housing, the team should include their names as the authors.

IMPORTANT The "Hardware" sheet is not required for simulation

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#	Component	Part name	Author	Source
1	Drive Motors	MG996R Servo 360° Rotation (Continuous)	TowerPro	makerstore.com.au
2	Long Range Distance Sensor	Adafruit VL53L4CX	Adafruit	adafruit.com
3	Short Range Distance Sensor	Adafruit VL53L4CD	Adafruit	adafruit.com
4	Small Servo Motors	FS90MR 9g Servo Motor	FeeTech	core-electronics.com.au
5	Microcontroller	Teensy 4.1	Teensy	core-electronics.com.au
6	Cameras	OpenMV AE3	OpenMV	openmv.io
7	GY-BNO055	9 axis IMU	No Brand	aliexpress.com
8	Small Switch	125V 6A Switch	E Switch	digikey.com.au
9	Large Switch	125V 10A Switch	E Switch	digikey.com.au
10	Colour Sensor	Adafruit AS7341 10 Channel Colour Sensor	Adafruit	adafruit.com
11	Battery	Nano-Tech 2S Cell 7.4V Battery	Turnigy power-systems	Pre Owned/ Unknown
12	Silicone for wheels	Transil 25A 2kg Silicone	Barnes	barnes.com.au
13	Ball Bearing	Ball Bearing - 8 mm ID, 22 mm OD 7 mm Race Width	RS Pro	au.rs-online.com
14	Fasteners			Pre Owned/ Unknown
15	Limit Switches	LS2501F350C2A	E-Switch	digikey.com.au
16	Needle Roller Bearing	17 mm ID Roller Bearing Needle, 30 mm OD	SKF	au.rs-online.com
17	Robot Frame	Custom Made	Ethan Seymour	CNC & 3D Printer
18	Wheels	Custom Made	Ethan Seymour	3D Printer & Silicone moulding
19	PCBs with electronic components	Custom Made	Ethan Seymour/ jlcpb	https://jlcpb.com/
20	Cables	Various Cables		core-electronics.com.au
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Name of the Local Currency	Total Cost Local Currency	Total Cost (\$) U.S.A. Dollars
Australian Dollar AUD	\$1,099.44	\$774.45

[illegible]

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#	Name	Software's Tool/Library	Description	Source
1	Visual Studio Code	Software's Tool	Software	code.visualstudio.com/
2	OpenMV IDE	Software's Tool	Camera Image Processing	openmv.io
3	Fusion	Software's Tool	3D CAD model & PCB Design	autodesk.com
4	Github	Software's Tool	Code Version Control	github.com
5	Bambu Studio	Software's Tool	3D Printing Slicer	bambulab.com
6	Makera Carvera Controller	Software's Tool	CNC Controller	global.makera.com
7	Edge Impulse	Software's Tool	AI model tool	www.edgeimpulse.com/
8	AS7341	Library	Colour Sensor Library	github.com/adafruit/Adafruit_AS7341
9	BNO005	Library	IMU Library	github.com/adafruit/Adafruit_BNO005
10	NeoPixel	Library	Neopixel RGB LED library	github.com/adafruit/adafruit_neopixel
11	VL53L4CD	Library	Short Distance LRF library	github.com/pololu/vl53l4cd-arduino
12	VL53L4CX	Library	Long Distance LRF library	github.com/stm32duino/VL53L4CX
13	Servo	Library	Servo Motor Library	github.com/michaelmargolis/SlowServo
14	BusIO	Library	I2C, UART, SPI control	github.com/adafruit/Adafruit_BusIO
15	Sensor	Library	Sensor drivers	github.com/adafruit/adafruit_sensor
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